

Moulya Vishwanath

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Cybersecurity & SRE-Focused Engineering Student | Cloud Monitoring | Python | AI Systems

PROFILE

Final-year Computer Science student with hands-on experience in security architecture, threat modeling, and cloud-connected backend systems. Currently designing a production-grade biometric authentication system at BitArmora. Published IEEE researcher with a filed patent in geofencing systems. Seeking to apply security and systems thinking in an SRE/DevOps environment focused on reliability and observability.

EDUCATION

SJB Institute of Technology, Bangalore
B.E. in Computer Science and Engineering

Expected 2026
CGPA: 8.79/10

Sadhana PU College, Karnataka
Karnataka Board

2022
Percentage: 84%

Sri Jnanakshi Vidyaniketan, Karnataka
CBSE Board

2020
Percentage: 80%

EXPERIENCE

Cybersecurity Analyst & Lead Architect
BitArmora

Feb 2026 – Present

- Led security architecture design for ArmoraBioKey, a biometric authentication system eliminating raw template storage across 3+ authentication workflows.
- Developed and configured GitLab CI/CD pipelines with SBOM generation, smoke tests, and automated test harness workflows to enhance security validation and operational reliability.
- Worked with Docker and Nexus Repository Manager to manage containerized artifacts and implement policy-driven validation workflows within CI/CD pipelines.
- Performed threat modeling and operational risk analysis, identifying and mitigating replay attack vectors in authentication pipelines.
- Researched and prototyped cryptographic template protection techniques including secure hashing to reduce biometric data exposure.
- Participated in incident analysis and mitigation planning for authentication-related operational risks.

Operational Intern
CRED

Nov 2025 – Jan 2026

- Contributed to a voice bot project for CRED Wallet, designing structured conversational workflows to improve loan-related query resolution.
- Built and maintained knowledge base content and automation workflows handling high-volume customer support tickets via Freshdesk.
- Configured API workflows and operational automation using Padawan, improving pipeline efficiency.

Trainee – Support
Tech Mahindra

June 2025 – July 2025

- Developed an HTML form integrated with backend servlets to automate author onboarding in Adobe Experience Manager (AEM).
- Gained hands-on experience with CMS workflows and operational user management in a client-facing environment.

RESEARCH & INNOVATION

AI-Driven Smart Waste Management with Geofencing and Civic Accountability

Published in IEEE

- Proposed an AI-based waste management system integrating dynamic geofencing, real-time location-aware alerts, and civic monitoring.
- Designed route optimization mechanisms and notification pipelines applicable to cloud-native alerting architectures.

Patent Application – System and Method for Dynamic Geofencing in Waste Management and Collection Notification

Field: Computer Science Domain

- Introduced a dynamic geofencing framework for automated, real-time boundary adaptation and alert triggering – analogous to threshold-based alerting in SRE monitoring systems.

PROJECTS

- **RegenX: Dynamic Geofencing-Based Waste Management Platform** | [GitHub](#) Ongoing
 - Built a cloud-connected real-time monitoring platform using Kotlin and Firebase Cloud Messaging, tracking complaint states and triggering location-based alerts via Google Maps API.
 - Developed an admin dashboard for real-time system observability, complaint monitoring, and status visualization – analogous to Grafana-style operational dashboards.
 - Integrated AI-assisted complaint validation using TensorFlow Lite, reducing false positives in alert generation.
- **Monitoring Dashboard Setup (Prometheus + Grafana)** 2026
 - Configured Prometheus locally to scrape and store system metrics (CPU, memory, disk) from a running Python application.
 - Built Grafana dashboards with threshold-based alert rules for infrastructure monitoring and proactive anomaly detection.
- **GoTogether: Smart Ride-Sharing & Rewards Platform** | [GitHub](#) March 2025 – April 2025
 - Developed a scalable Node.js/Express backend with MongoDB supporting real-time route optimization and payment integration.
 - Designed REST APIs for ride coordination, tested and validated via Postman across multiple concurrent user flows.
- **CogniChat: Cognitive Assessment Therapist** | [GitHub](#) Dec 2024
 - Built a Flask/MongoDB chatbot using GPT and BERT with dynamic persona switching and emotion-aware conversation flows.
 - Designed, documented, and validated REST API endpoints using Postman for reliable backend integration.

TECHNICAL SKILLS

Languages: Python (Advanced), JavaScript, Kotlin, C, Java, Bash (Basic)

Cloud & Databases: Firebase, SQL, MongoDB, AWS Fundamentals (EC2, CloudWatch concepts)

Monitoring & Observability: Prometheus (local setup), Grafana (dashboards & alerts), Firebase Cloud Messaging

Security: Threat Modeling, Cryptographic Hashing, Biometric Template Protection, Secure Workflow Design

Tools & Frameworks: Git, GitHub, Node.js, Express.js, Flask, React.js, Android Studio, Postman, Freshdesk

AI/NLP: TensorFlow Lite, BERT, GPT integrations

Soft Skills: Leadership, Teamwork, Adaptability, Time Management, Critical Thinking

EXTRACURRICULAR

- Finalist / Top 20 in multiple national-level hackathons: SIET Tumkur National Level, SKIT 'Build for Bengaluru', and 24-hour hackathon at BGSCET.
- Organized social media awareness campaign for school students on digital safety.